User Guide

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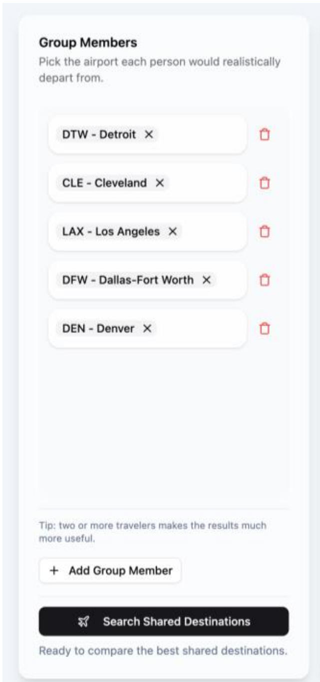
Group Members



A group member represents one person that will be traveling to the destination. Each group member has a list of airports (can be just one) that they are able to fly out of. There must be at least 2 group members present in order to find a destination.

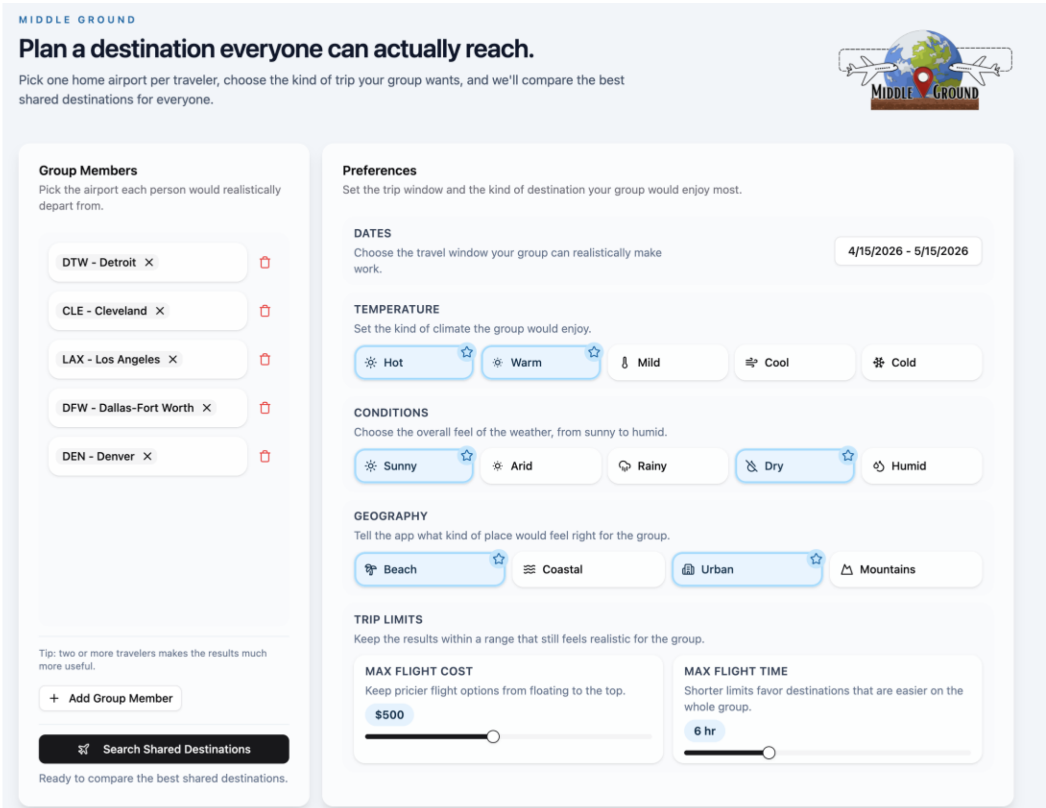
To add a new group member, click the “Add Group Member" button.

To remove a group member, click the red trashcan icon next to the group member.

To add an airport to a group member, simply start typing the city name, airport name, or airport code into the search box. As you type, a list of airports matching your query will be displayed. Click on one of them to add it or press “Enter” to add the currently selected airport.

To remove an airport from a group member, simply click on the X.

Preferences



Dates

The "Dates" section allows you to specify a range of dates during which you want your flights to take place. Click the button to open the calendar; then, select the start and end dates. The start date should be the **earliest** possible date that everyone in the group can fly out on, and the end date should be the **latest** possible date that everyone in the group can return home.

Weather

The "Weather" section allows you to set preferences of what sort of climate you would like to have at the destination. Clicking a preference option does not guarantee that condition in the destination, but it gives preference towards it in the search process. You can select multiple options within the same category.

The Temperature section has the following options:

- Hot (>85°F)
- Warm (70°-85°F)
- Mild (55°-70°F)
- Cool (40°-55°F)
- Cold (<40°F)

These options filter destination locations based on their average temperature. More than one filter can be selected if desired.

The Conditions section has the following options:

- Sunny
- Arid (Low precipitation)
- Rainy (High precipitation)
- Dry (Low humidity)
- Humid (High humidity)

These options filter destination locations based on their average weather conditions. Similar to temperature, multiple can be selected.

Geography

The "Geography" section allows you to set preferences based on the geography of potential destination locations. Like Weather preferences, these preferences are not guaranteed but help to filter potential destination options. You can select multiple options within the same category.

The Environment section has the following options:

- Beach
- Coastal
- Urban
- Mountains

Airport Searching

Once you have finalized your preferences, the user will hit the "Search Shared Destinations" button at the bottom of the screen. The information the user typed in for each group member and the attributes previously explained above will be sent to the backend server. On the server, this search will be saved in a database to increase program efficiency and reduce redundancy if this search is made again.

After this information is stored in the backend database, the FlightAPI then takes this data to be used in calculating the best airport for your group's land and meet up together. At the end of the filtering process, the user receives airports that are ranked by percentages based on what they are looking for. The ranking process is as follows.

Ranking

The ranking portion is split into two processes: the airport ranking and the flight ranking. We start off with the airport ranking and then proceed to the flight ranking after.

The airport ranking starts by taking in the temperature, conditions, and geography from the UI and runs those attributes against all airports. As previously mentioned, each individual condition will get a ranking from 0 to 1 between what's desired (inputs on the UI) and the actual data corresponding to each airport which is stored in our local database. The overall airport score is the average of all the individual scores. From there we go to the flight ranking.

The flight ranking takes the top 5 airports previously ranked and tries to get flight details from each of the origin airports to those airports. If an origin airport can’t reach one of the ranked airports (i.e. Anchorage, Alaska as an origin and Orlando as a top airport), that airport is skipped and it moves on to the next top. This repeats until there are 5 reachable airports or until there aren’t any more airports left. Once we have the top 5 airports and the flights between the origin airports and those, we rank the flights based on the rest of the inputted data like date ranges, flight cost, and flight duration. We take the overall flight score from each of the routes and then average them for the final flight score for each destination airport.

Now, we have an airport score and a flight score for each of the top 5 airports that match all conditions. Those two scores are averaged to get an overall score for the destination. These are put in order from highest to lowest and if there is a tie, then the cheaper trip comes first.